

PPE Compliance Check Agent

Tier 3 Autonomous AI Agent for PPE Safety Monitoring

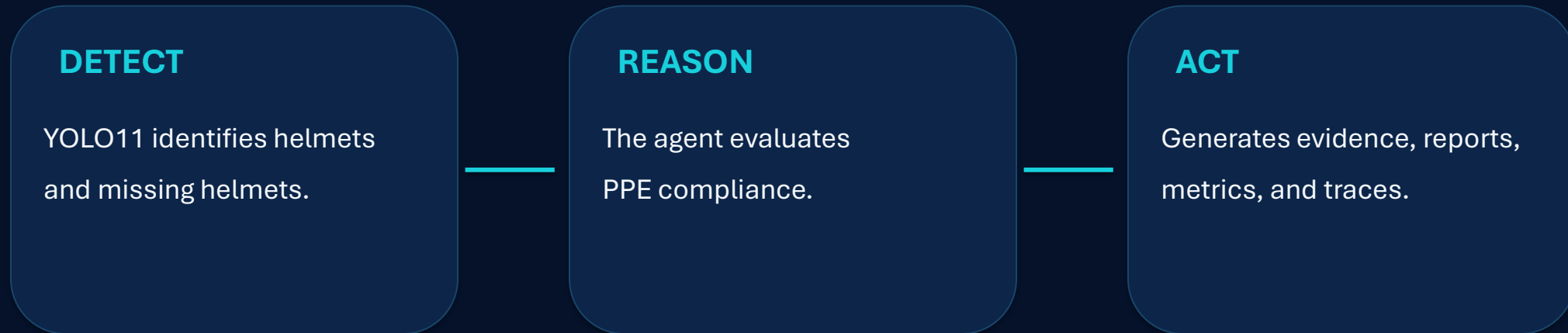
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Project Overview

A Tier 3 autonomous computer vision agent

Goal: Automate PPE helmet compliance monitoring in construction and industrial environments.



Project Tier: Tier 3 – Custom-Trained CV Agent

Tier 3 qualification: custom-trained YOLO11 PPE detection model

Problem & Solution

PROBLEM

Manual PPE monitoring can be:

- Time-consuming
- Inconsistent
- Difficult to scale
- Dependent on continuous human observation

SOLUTION

PPE Compliance Check Agent:

- Detects helmet use automatically
- Identifies potential violations
- Handles uncertain cases
- Generates documented evidence
- Supports image and video monitoring

Computer Vision + AI Reasoning → Automated Safety Monitoring

Agent Architecture

Perception → Reasoning → Action

1. Input Ingestion

3. Preprocessing

5. Compliance Reasoning

7. Trace Logging

9. Continuous Monitoring

2. Input Validation

4. YOLO11 Perception

6. Action & Evidence

8. Evaluation

End-to-End Agent Pipeline

Dataset & YOLO11 Model

MODEL & DATA

- Custom PPE dataset
- Training, validation, and evaluation
- Custom-trained YOLO11 detector
- Helmet / no-helmet conditions
- Image and video inference
- Representative samples stored in GitHub



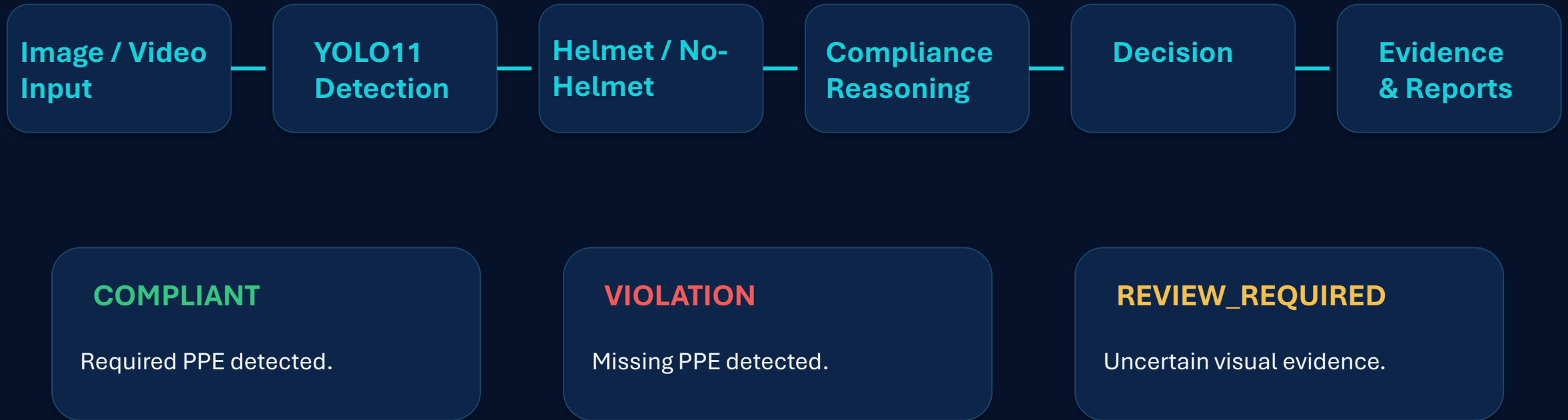
PPE
Dataset

YOLO11

Detector

Detection

How the Agent Works



Outputs: Annotated Images • Videos • JSON Reports • Metrics • Execution Traces

PPE Compliance Results

Three decision outcomes from the agent

COMPLIANT

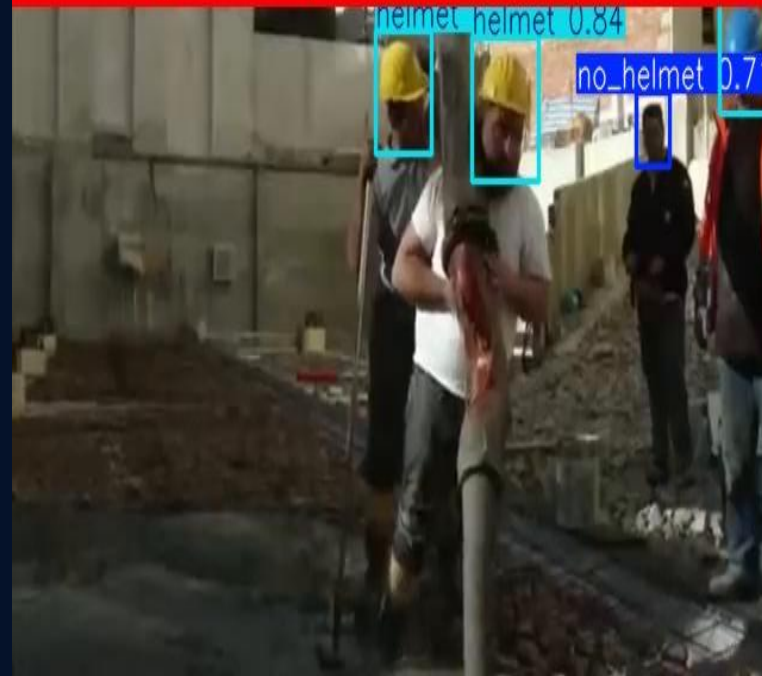
Scenario 7 | COMPLIANT | Frame 1397



Required PPE detected.

VIOLATION

Scenario 4 | VIOLATION | Frame 698



Missing PPE detected.

REVIEW_REQUIRED

Scenario 3 | REVIEW_REQUIRED | Frame 465



Insufficient or uncertain evidence.

Evaluation & Metrics

10 representative scenarios

COMPONENT-LEVEL

- Object detection performance
- Inference latency
- Detection consistency
- Annotated output quality

KEY MODEL RESULTS

Precision: **0.9147**

Recall: **0.8911**

mAP50: **0.9285**

mAP50-95: **0.6053**

Inference: **3.7 ms/image**

SYSTEM-LEVEL

- Successful image/video processing
- PPE compliance classification
- Annotated evidence generation
- Structured report generation
- Execution trace generation

AGENT EVALUATION

10 scenarios evaluated

6 COMPLIANT

2 VIOLATION

2 REVIEW_REQUIRED

100% processing success

Average agent inference time: **11.71 ms**

Artifacts: Evaluation Images • Monitoring Videos • JSON Reports • Metrics • Agent Traces

Failure Case Analysis

Why REVIEW_REQUIRED matters

CASE 1:

Limited PPE Evidence

↓

No reliable helmet / no-helmet detection

↓

REVIEW_REQUIRED

CASE 2:

No Relevant Worker Visible

↓

PPE status cannot be evaluated

↓

REVIEW_REQUIRED

Scenario 10 | REVIEW_REQUIRED | Frame 2096



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Key takeaway:

The agent avoids forcing a compliance decision when the visual evidence is uncertain.

GitHub & Reproducibility

Repository Structure

data/
docs/
models/
notebooks/
results/
README.md
requirements.txt
run_agent.py

The screenshot shows the GitHub repository page for 'PPE-Compliance-Check-Agent' by user 'Evaabou20'. The repository is public and has 1 branch (main) and 0 tags. The file list shows the following structure:

File/Folder	Description	Last Commit
data	Complete final project documentation	2 days ago
docs	Update AI_usage_log.md	1 hour ago
models	Add pretrained model download instructions	4 hours ago
notebooks	Update README.md with final project notebook details	2 hours ago
results	Final PPE Compliance Check Agent Project	2 days ago
README.md	Enhance README with execution command and verification ...	14 minutes ago
requirements.txt	Add gdown dependency for model download	25 minutes ago
run_agent.py	Add single-command PPE agent runner	28 minutes ago

The right sidebar contains the 'About' section, which describes the project as an 'AI-powered computer vision agent for automated PPE compliance monitoring using YOLO11'. It also shows 'Releases' (No releases published) and 'Packages'.

Single-command execution: `python run_agent.py`

Includes: final notebook • sample data • results • metrics • reports • traces • documentation

Key Learnings & Future Improvements

KEY LEARNINGS

- Train and evaluate a custom YOLO11 detector
- Build an end-to-end computer vision agent
- Combine perception, reasoning, and action
- Evaluate AI systems systematically
- Document projects for reproducibility

FUTURE IMPROVEMENTS

- Detect vests, gloves, goggles, and boots
- Integrate live surveillance cameras
- Deploy as a real-time application
- Improve robustness to occlusion and lighting
- Expand reasoning capabilities

PPE Compliance Check Agent

Computer Vision + Reasoning + Evaluation

for Automated PPE Safety Monitoring

DETECT

Automated PPE detection with YOLO11

DECIDE

Intelligent compliance decision-making

DOCUMENT

Evidence, reports, metrics, and traces

Thank You!